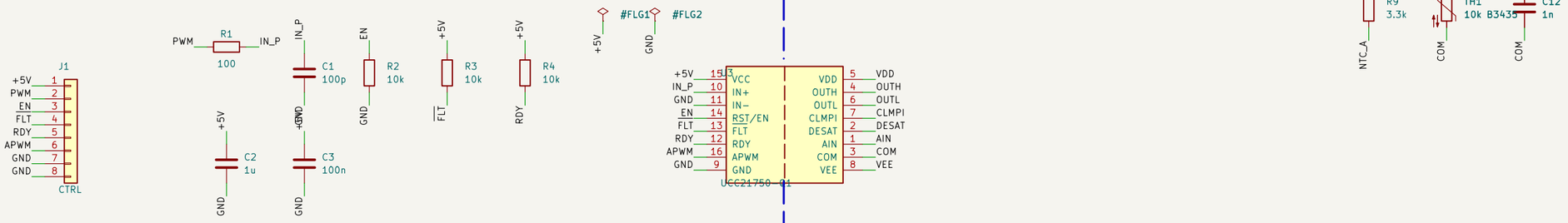
LOW-VOLTAGE SIDE (vehicle 12 V domain, GND)

UCC21750-Q1 isolated gate driver

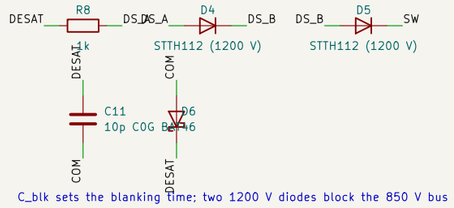
Temperature: 200 uA from AIN into an NTC on the cold plate



J1: +5V, PWM, EN, FLT (open drain), RDY, APWM (temperature), GND

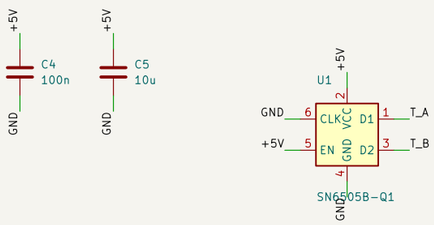
RC deglitch on PWM; EN pulled low = output off if the controller resets

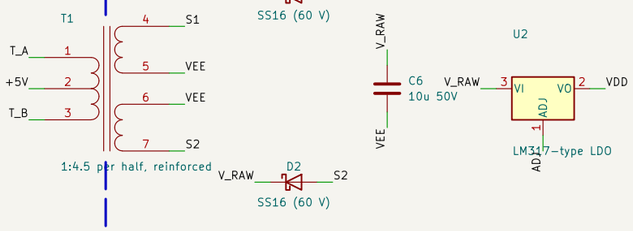
DESAT: trips when Vds stays > ~7.9 V with the gate on



C_blk sets the blanking time; two 1200 V diodes block the 850 V bus

ISOLATED BIAS SUPPLY: push-pull transformer, 19 V regulated, split into +15 V / −3.9 V



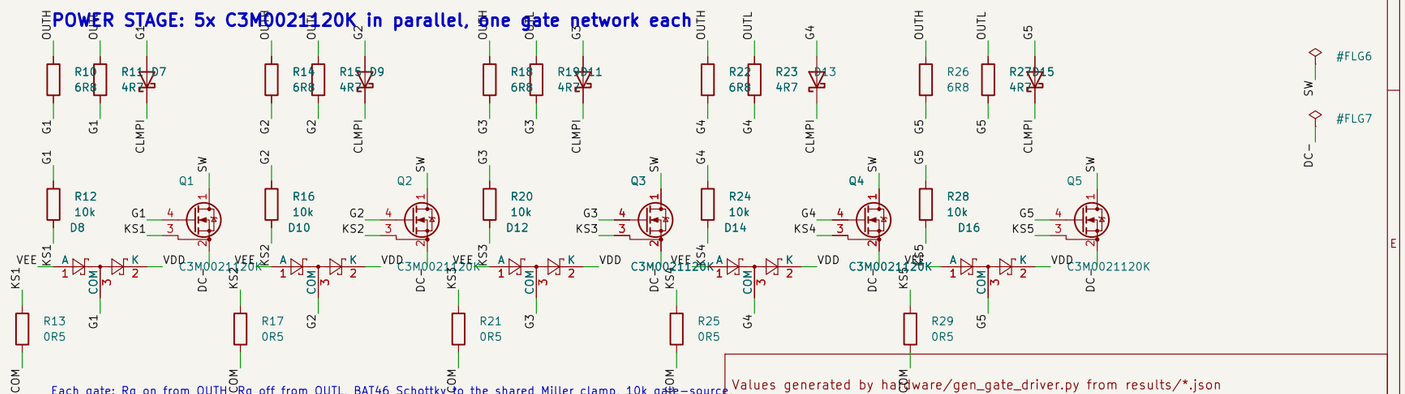


10 uF + 100 nF per rail at the driver pins (datasheet: > 10 uF)

VDD − VEE = 1.25 V x (1 + 3.4k / 240) = 19.0 V (LM317-type automotive LDO)

3.9 V Zener (BZT52C3V9-Q) sets COM above VEE: VDD = +15.1 V, VEE = −3.9 V vs source

POWER STAGE: 5x C3M0021120K in parallel, one gate network each



Each gate: Rg_on from OUTH, Rg_off from OUTL, BAT46 Schottky to the shared Miller clamp, 10k gate-source.

BAT54S clamps the gate between VEE and VDD, 0.5 ohm Kelvin-source resistor

Design notes (all values from the simulation study in this repo):

- gate charge 159 nC per device, drive power 0.30 W per switch at 20 kHz
- isolated supply 2 W (margin 5.3x); peak source/sink 4.2 / 10.0 A
- DESAT delay 0.72 us typ / 0.96 us worst; CMTI margin 1.7x
- keep HV nets (SW, DC−, DS_B) >= 8 mm from the LV side; slot the PCB under U1 and T1

ISOLATION BARRIER (reinforced)

Values generated by hardware/gen_gate_driver.py from results/*.json
Rg_on = 6.8 ohm, Rg_off = 4.7 ohm per device; C_blk = 10 pF (from simulation)
800 V SiC traction inverter — design study

Sheet: /
File: gate_driver.kicad_sch

Title: Isolated gate driver — one switch (5x SiC MOSFET in parallel)

Size: A3	Date: 2026-10-07	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/1